JARETT POLINER
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Research engineer at Los Alamos National Laboratory working at the intersection of AI, simulation, and complex physical systems. Develops generative models, learned simulation systems, and multimodal world models for dynamic, partially observed environments, with an emphasis on rigorous evaluation and scalable experimentation.

EXPERIENCE

Los Alamos National Laboratory, Earth and Environmental Sciences (LANL, EES) Los Alamos, NM
Postdoctoral Research Associate October 2025–Present

- Build real-to-sim pipelines that turn partial micro-CT scans into simulation-ready 3D particle environments, combining variable-size graph autoencoding of particle connectivity with latent diffusion over particle geometry to eliminate manual asset construction.
- Develop recursive void-conditioned packing for multiscale super-resolution of generated structures, enforcing physical feasibility (no interpenetration of contacting grains) and realistic size and shape distributions at generation time.
- Evaluate generated environments against micro-CT ground truth and downstream physics simulations, testing both geometric validity and preservation of structure–property behavior.
- Contribute to Transparent Earth, a multimodal subsurface foundation model; investigate context scaling through query-conditioned observation selection and learned retrieval.

Multimodal Urban Traffic World Model — Columbia University and Creative Mobile Technologies (CMT)
Collaborator January 2026–Present

- Develop a flow-matching world model for a dynamic, partially observed urban environment, learning network-scale state dynamics from sparse moving-agent telemetry, fixed cameras, road-network topology, and event context.
- Reconstruct and forecast latent system state from incomplete, irregularly sampled observations, combining graph-based state estimation with learned temporal dynamics for long-horizon prediction.
- Design controlled evaluations that systematically vary observability, sensor availability, and external context to quantify model capability, failure modes, and information requirements at network scale.
- Build reproducible data and experiment infrastructure for heterogeneous asynchronous sensor streams, including ingestion, dataset construction, experiment orchestration, and large-scale evaluation.

Columbia University, Civil Engineering Department New York, NY
Doctoral Research Assistant September 2019–August 2025

- Replaced the local return-mapping iteration in a mesh plasticity solver with a learned projection, removing a per-material-point nonlinear solve and yielding order-of-magnitude speedups on large-scale finite-element benchmarks, while prioritizing solver stability and throughput alongside accuracy (CMAME 2025, co-first author).
- Designed an active-learning system for expensive simulation environments, reconstructing high-dimensional response surfaces in fewer than 30 queries using Bayesian optimization, maximum-entropy sampling, and conditional latent diffusion.
- Generated simulation-ready 3D geometry with graph (GraphRNN) and conditional diffusion models, producing microstructures whose topology and morphology match micro-CT references and that feed directly into downstream solvers.

Air Force Research Laboratory, Munitions Directorate (AFRL, RW) Eglin Air Force Base, FL
AFRL Scholar June 2023–August 2023

- Applied generative AI, graph convolutional neural networks, and Fourier shape analysis to predict void shapes for prescribed critical threshold velocity during pore-collapse scenarios.
- Presented research outcomes to Air Force personnel and contributed to an internal research paper on data augmentation and generative prediction methods for energetic materials.

Columbia University, Civil Engineering Department New York, NY
Graduate & Undergraduate Research Assistant May 2018–December 2019

- Developed finite-element simulation workflows for pediatric heart-valve prosthetics and computational workflows for 3D concrete printing.
- Wrote and optimized scripts for 3D concrete printing, systematically testing extrusion rates, infill patterns, and speed.

SELECTED PUBLICATIONS AND PREPRINTS

Published

- Xiao, M.*, Poliner, J.*, & Sun, W. C. (2025). Geometric learning for computational mechanics Part IV: Efficient mesh-based plasticity from a domain-specific foundation model. *Computer Methods in Applied Mechanics and Engineering*, 446, 118310.
  *Co-first authors / contributed equally.
- Poliner, J., Sun, W. C., Alshibli, K. A., & Regueiro, R. A. (2026). Deep generative graphs for synthesizing microstructure of topology-preserved polymer-bonded explosives. *Engineering with Computers, in press.* https://doi.org/10.1007/s00366-026-02445-8
- Poliner, J., Crone, J. C., Bryant, E. C., Bennett, K. C., & Sun, W. C. (2026). Conditional diffusion-based design of experiments for yield surface discovery with limited experimental data. *Scientific Reports, in press.*

Under Review / In Preparation

- Poliner, J., & Di, X. (2026). FleetNet: Learning Urban Traffic State Dynamics Under Partial Observability and Disruptions. *Transportation Research - Part C, In preparation*.
- Doan, J.*, Poliner, J.*, Bryant, E. C., Bennett, K. C., Miller, N. A., & Hill, R. G. (2026). Variational graph autoencoding for latent compression and reconstruction of granular microstructures. *In preparation.*
  *Co-first authors
- Poliner, J., Bryant, E. C., Bennett, K. C., & Miller, N. A. (2026). Particle Packing Super-Resolution: Recursive Void-Conditioned Generation of Multiscale Granular Microstructures. *In preparation.*

SKILLS

Machine Learning: PyTorch, PyTorch Geometric, transformers, graph neural networks, diffusion models, flow matching, generative modeling, representation learning, multimodal systems, world models, evaluation and benchmarking
Simulation & Evaluation: learned simulation, partially observed systems, dynamic environments, state estimation, environment evaluation, active experimental design
ML Systems & Infrastructure: Python, Linux, Git, Docker, containerized execution, HPC clusters, GPU computing, CUDA, experiment orchestration, large-scale data pipelines
Scientific Computing: finite-element and discrete-element methods, numerical optimization, inverse problems, solver stability
Programming: Python, MATLAB, SQL
Simulation Tools: NVIDIA Warp, ABAQUS, Yade
Clearance: Active U.S. Department of Energy (DOE) Q Clearance

EDUCATION

Columbia University, Fu Foundation School of Engineering and Applied Science
Ph.D., *Civil Engineering with a concentration in Computational Mechanics,* August 2025
Dissertation: *Generative Modeling with Sparse Data for Solid Mechanics*
Awards: 2023 AFRL Scholarship, 2022 DoD NDSEG Poster Award, 2020 NDSEG Fellowship, 2020 Guggenheim Fellowship

M.S., *Civil Engineering with a concentration in Computational Mechanics,* May 2020

B.S., *Engineering Mechanics,* May 2019
Minors: Applied Mathematics, Civil Engineering and East Asian Studies

Honors: 2018 & 2019 Dean's List, 2018 Concrete Institute Foundation (CIF) Scholarship

ATHLETICS AND LEADERSHIP

Columbia University, NCAA Division I Varsity Men's Fencing Team New York, NY
Epee Fencer (Captain, 2018–2019); 2016 and 2019 NCAA Team National Champions August 2015–2019